MIL-C-48060 (PA)
27 November 1973
SUPERSEDING
(See 6.5)

MILITARY SPECIFICATION

CARTRIDGE, 105MM, HE: M444E1,

FUSIBLE PLUG ASSEMBLY FOR

This Specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the parts for one type of plug designated as Plug Assembly, Fusible.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-A-2550	- Ammunition, General Specification for

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109	- Quality Assurance Terms and Definitions

FSC: 1315

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MIL-STD-1168 - Lot Numbering of Ammunition

MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components

MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

U.S. Army Armament Command

D-9247311 - Fusible Plug Assembly

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with the applicable drawings and specifications.

3.2 Fusible Plug.-The Fusible Plug shall comply with all requirements specified on Drawing (Dwg) D-9247311, all associated drawings, and with all requirements specified in applicable specifications.

3.3 Air Pressure Test.-There shall be no evidence of leakage when the fusible plug is subjected to an air pressure of 40 psig, minimum, for 15 seconds, minimum. The test shall be performed as specified in 4.4.2.

3.4 Workmanship.-All parts and assemblies shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, corrosion products, and other foreign matter. The cleaning method used shall not be injurious to any parts, nor shall the parts be contaminated by the cleaning agents. All marking and stamping shall be neat and sharply defined.

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3.5 Submission of results of contractor-conducted examinations and tests.-The contractor shall forward copies of the results of any of his examinations or tests to the designated address as required by the contract. (see 6.4).

3.6 First article testing.-This specification makes provisions for first article testing. Submission of first article quantity by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspection set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all the quality assurance provisions specified in this specification.
b. Drawing and specification number and date, together with identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements when such material is purchased by the contractor or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.
d. Number of items in the lot.
e. Date submitted.

The certificates shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind

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his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection.

4.2.1 Submission.-The contractor shall submit a first article quantity as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.2.2. (See 6.1.) The first article sample shall consit of the following:

a. Fifty (50) fusible plug assembly: D-9247311
b. Twenty-five (25) plug: D-9248618
c. One (1) pound of fusible alloy

All parts and completed assemblies submitted shall have been produced by the Contractor (or furnished by a supplier) using the same production processes, procedures and equipment as will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. The sample shall be accompanied by certificates of conformance. A first article quantity, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing specification or source of supply as to significantly affect product uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer.

4.2.2 Inspection to be performed.-All parts and assemblies shall be subjected by the Government to any or all the examinations or tests specified in this specification and any or all requirements of the applicable drawings. In addition, the following tests shall be performed:

4.2.2.1 Air pressure test.-The fifty (50) fusible plug assemblies shall be subjected to the air pressure test specified in 4.4.2.

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4.2.2.2 Push test of ternary eutectic alloy.-After completion of the air pressure test, twenty-five (25) of the fusible plug assemblies shall be subjected to the push-out test specified in 4.4.3.

4.2.2.3 Melting test of ternary eutectic alloy.-One-quarter (1/4) pound of the fusible alloy shall be subjected to the melting test specified in 4.4.1.

4.2.3 Rejection.-If any part or assembly fails to comply with any of the applicable requirements, the first article quantity shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any part or assembly, in the sample to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article quantity or portion thereof. Until a first article quantity is accepted, the contractor is in no way authorized by the Government to resume regular production unless otherwise directed by the Contracting Officer.

4.3 Inspection provisions.-

4.3.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process in accordance with the same drawing, same drawing revision, same specification and same specification revision, and complies with the provisions for submission of product as specified in MIL-STD-105. Change to process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot shall contain:

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a. Ternary Eutectic Alloy from one lot number.
b. Plugs of one lot interfix from one manufacturer.

4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plan for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Individual defects shall have an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Plug (see Dwg. D-9248618).

Categories	Defects	Method of Inspection	Code No. (See 6.2)
Critical:	None defined		
Major:			
101.	Length of plug	Gage	01001
102.	Diameter of hole	Gage	01002
103.	Distance between slots	Gage	01003
104.	Location of slots	Gage	01004
105.	Width of slots	Gage	01005
106.	Distance from bottom of flange to slot	Gage	01006
107.	Distance across parallel sides of hexagon	Gage	01007
108.	Pitch diameter of thread	Gage	01008
109.	Major diameter of thread	Gage	01009
110.	Outside diameter	Gage	01010
Minor:			
201.	Distance from end of threads to bottom of flange	Gage	01011
202.	Distance from end of plug to bottom of flange	Gage	01012
203.	Surface finish improper on any surface	Visual	01013
204.	Taper missing (when applicable)	Visual	01014
205.	Evidence of poor workmanship (See 3.4)	Visual	01015

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4.3.2.2 Fusible plug assembly, prior to pouring alloy (See Dwg. D-9247311).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major			
101.	Primer or sealing compound not present on noted surface	Visual	02001
Minor:	None defined		

4.3.2.3 Fusible plug assembly, after pouring alloy, (See Dwg. D-9247311).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major			
101.	Alloy excessively below or excessively above noted surface	Gage	03001
Minor:			
201.	Evidence of poor workmanship (See 3.4)	Visual	03002

4.3.2.4 Container, sealed (See Section 5).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major			
101.	Container damaged	Visual	04001
102.	Contents of container loose	Visual/Manual	04002
103.	Container incorrect according to size and weight	Visual	04003
104.	Closing of container incorrect or incomplete	Visual	04004
105.	Marking data missing, illegible, or incomplete	Visual	04005
Minor			
201.	Size of marking letters incorrect	Visual	04006

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4.3.3 Testing.

4.3.3.1 Melting test of ternary eutectic alloy, (See Dwg. D-9247311) Major Defect, Code No. 05001.-Five (5) samples of approximately one-quarter (1/4) pound each shall be randomly taken from each lot of ternary eutectic alloy to be used in each lot of fusible plug assemblies. If any sample fails to comply with the melting point specified on the applicable drawing, the alloy lot shall be rejected. In addition, all fusible plug assemblies containing the rejected lot of ternary eutectic alloy shall be rejected. The test shall be conducted as specified in 4.4.1 using equipment in accordance with 4.3.4.

4.3.3.2 Air pressure test of fusible plug assembly (See Dwg. D-9247311 and paragraph 3.3) Major Defect, Code No. 06001.-Every plug shall be subjected to this test. If any part fails to comply with the applicable requirement, it shall be classed defective and removed from the lot. The test shall be performed as specified in 4.4.2 using equipment in accordance with 4.3.4.

4.3.3.3 Push-out test of ternary eutectic alloy (See Dwg. D-9247311) Major Defect, Code No. 07001.-Ten plugs shall be selected from each lot after the air pressure test has been performed. If any plug fails to comply with the requirement specified on the applicable drawing, the lot represented by these parts shall be rejected. The test shall be performed as specified in 4.4.3 using equipment in accordance with 4.3.4.

4.3.4 Inspection equipment.-The inspection equipment required to perform the examinations and tests prescribed in this section is identified under the "Method of Inspection" headings in 4.3.2 and in 4.3.3 (See 6.3).

4.4 Test methods and procedures.

4.4.1 Melting test of ternary eutectic alloy.-The samples selected in 4.3.3.1 shall be individually tested by placing each in a beaker of distilled water at ambient temperature. The temperature of the water shall then be raised at a rate not to exceed 10°F per minute. The water shall be stirred to assure uniform heating. A temperature-indicating device such as a therometer or themocouple, suitable for the required temperature range, shall be inserted in the water. When the alloy is observed to melt, the indicated temperature shall be recorded. The measured temperature shall then be compared with the value specified on the applicable drawing to determine compliance or non-compliance of the sample with the melting point requirement.

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4.4.2 Air pressure test of Fusible Plug Assembly.-Each fusible plug assembly specified in 4.3.3.2 shall be subjected to this test. The test shall be performed by sealing the open end of the plug and completely immersing the plug in water. The air pressure specified on the applicable drawing shall then be applied through the sealed end of the plug for the specified time intended. Observation shall be made for air bubbles originating from the area of the fusible alloy indicating a leaking plug. All such leaking plugs shall be removed from the lot. All equipment used for this test shall be approved in accordance with 4.3.4.

4.4.3 Push test of Ternary Eutectic Alloy.-Each fusible plug assembly selected in 4.3.3.3 shall be mounted in an approved fixture and the load specified on the applicable drawing shall be applied in the direction shown. Rate of loading shall not exceed 1000 pounds per minute. When the load has been applied the fusible alloy shall be checked against the applicable requirement. Non-complying assemblies shall be rejected. All equipment and fixturing used for this test shall be approved in accordance with 4.3.4. Fusible plug assemblies subjected to this test shall not be returned to any lot. (Destructive test)

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level C.-The fusible plug assemblies shall be preserved and packaged in accordance with MIL-STD-1169.

5.2 Packing.

5.2.1 Level C.-The fusible plug assemblies, preserved and packaged as specified in 5.1.1, shall be packed for shipment in accordance with MIL-STD-1169.

5.3 Marking.-Marking of the carton shall comply with MIL-STD-1169.

5.4 Shipping.-When components from more than one lot are shipped as a carload, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

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a. Title, number and date of this specification.
b. Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.
c. Provisions for submission of first article samples.

6.2 Inspection code numbers.-The five-digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection Equipment Designs.-Inspection equipment designs are of two types: Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers under the "Method of Inspection" headings in Section 4. Design responsibility for all other inspection equipment is assigned to the contractor. The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.3.2 and 6.3.3 should he elect to do so.

6.3.1 SIE designs.-SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.
b. A source control drawing or a specification control drawing as defined in MIL-STD-100.
c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detail design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.3.2 Contractor designs.-Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is catalogued and available for purchase by the general public). Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which

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provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with SIE designs if not otherwise covered thereby (see 6.3.1c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirement Lists (DD Form 1423).

The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification number and the applicable five-digit defect code number (or other specific identifying information) from Section 4 of this specification shall be referenced on each contractor design.

6.3.3 Submission of designs for approval.-Contractor designs shall be approved by the Government prior to fabrication or procuring the equipment. Designs shall be submitted for approval in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423). Partial submission of inspection equipment date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt.

6.4 Submission of results of contractor-conducted examinations and tests.-Unless otherwise specified by the Contracting Officer, the contractor shall forward requested records of examinations or tests to Commanding Officer, Picatinny Arsenal, Attn: SARPA-QA-A-S.

6.5 Supersession data.-This specification includes the requirmeents of, and supersedes, Purchase Specification AEI-513, dated 22 September 1969.

6.6 Distribution of ammunition data cards.-Distribution of data cards shall include the following: Commander, Picatinny Arsenal, Attn: SARPA-QA-A-S, Dover, New Jersey 07801.

Custodian:
Army-PA

Preparing Activity:
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Project Number: 1315-A013

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